

Math GRE Prep Day 2

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

- 1 Calculus Problems
 - Standard problems
 - Somewhat trickier problems

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Problem 11

Evaluate

$$\lim_{n \rightarrow \infty} \left(\sqrt{1 + 2 + 3 + \cdots + n} - \sqrt{1 + 2 + 3 + \cdots + (n-1)} \right)$$

Problem 12

Evaluate

$$\lim_{n \rightarrow \infty} \frac{\sqrt{1 \cdot 2} + \sqrt{2 \cdot 3} + \cdots + \sqrt{n(n+1)}}{n^2}$$

Problem 13

Evaluate

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 14

Evaluate

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 15

If $f(x) = e^{x^3} + e^{-x^3}$, evaluate $f^{(2022)}(0)$

Problem 16

Calculate

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{\sqrt{1^3 + 2^3 + \dots + i^3}}$$

Problem 17

Calculate

$$\lim_{x \rightarrow \infty} (\sin \sqrt{x + \pi} - \sin \sqrt{x})$$

Problem 18

Calculate

$$\lim_{n \rightarrow \infty} \cos \frac{1}{2} \cos \frac{1}{4} \cos \frac{1}{8} \dots \cos \frac{1}{2^n}$$